CSC-125 Python Programming
Midterm Exam Practice Questions

Covers Weeks 1-9: Variables, Data Types, Strings, Operators, Conditionals, Lists, Tuples, and Loops

Name: _____ **Date:** _____

Instructions:

- This practice exam contains 50 questions similar to the actual midterm
- The actual midterm will be on paper with no computers, phones, or notes allowed

Part 1: Multiple Choice

1. What data type is "42"?

a) integer
b) string
c) float
d) boolean

2. Which is a valid variable name?

a) 3total
b) total_score
c) total-score
d) total score

3. What does input() always return?

a) integer
b) string
c) float
d) depends on user input

4. What is the output of: print(5 + 3 * 2)?

a) 16
b) 11
c) 13
d) Error

5. What does "python"[0] return?

a) "p"
b) "P"
c) 0
d) Error

6. What is the result of: 10 % 3?

a) 3
b) 1
c) 0
d) 3.33

7. What does len([10, 20, 30]) return?

a) 2
b) 3
c) 60
d) Error

8. Which creates a tuple?

a) [1, 2, 3]
b) (1, 2, 3)
c) {1, 2, 3}
d) <1, 2, 3>

9. What is the output of: print(7 // 2)?

a) 3.5
b) 3
c) 4
d) 3.0

10. What does "Hello" + "World" produce?

a) "Hello World"
b) "HelloWorld"
c) Error
d) "Hello + World"

11. Given x = 5 and y = 10, what is x < y?

a) True
b) False
c) 5
d) 10

12. What is the result of: True or False?

a) True
b) False
c) Error
d) None

13. What will print?

```
age = 20
if age >= 18:
    print("Adult")
```

a) "Adult"
b) Nothing
c) Error
d) 20

14. Given numbers = [5, 10, 15], what is numbers[2]?

a) 5
b) 10
c) 15
d) Error

15. What does range(5) produce?

a) [1, 2, 3, 4, 5]
b) [0, 1, 2, 3, 4]
c) [0, 1, 2, 3, 4, 5]
d) 5

16. What is the output?

```
for i in range(3):
    print(i)
```

a) 1 2 3
b) 0 1 2
c) 0 1 2 3
d) 3

17. Given text = "Python", what is text[-1]?

a) "P"
b) "n"
c) "o"
d) Error

18. What does "hello".upper() return?

a) "Hello"
b) "HELLO"
c) "hello"
d) Error

19. Which comparison operator means "not equal"?

a) !=
b) ==
c) <>
d) =/=

20. What is the output of: print(not False)?

a) True
b) False
c) 0
d) 1

21. What does "Python"[1:4] return?

a) "Pyt"
b) "yth"
c) "ytho"
d) Error

22. What is the output?

```
score = 75
if score >= 90:
    print("A")
elif score >= 80:
    print("B")
elif score >= 70:
    print("C")
else:
    print("F")
```

a) "A"
b) "B"
c) "C"
d) "F"

23. Given fruits = ["apple", "banana"], what does fruits.append("cherry") return?

a) ["apple", "banana", "cherry"]
b) None
c) 3
d) "cherry"

24. What is the result of: True and False?

a) True
b) False
c) Error
d) None

25. What does [1, 2, 3] * 2 produce?

a) [2, 4, 6]
b) [1, 2, 3, 1, 2, 3]
c) [1, 2, 3, 2]
d) Error

Part 2: Fill in the Code

26. Convert this string to an integer:

```
age_text = "25"
age_number = _______(age_text)
```

27. Create a boolean variable set to True:

```
is_valid = _______
```

28. Get the length of a list:

```
scores = [85, 90, 78]
count = _______(scores)
```

29. Add an item to the end of a list:

```
fruits = ["apple", "banana"]
fruits._______("cherry")
```

30. Access the last element using negative indexing:

```
numbers = [10, 20, 30, 40]
last = numbers[_______]
```

31. Create a for loop that runs 5 times:

```
for i in _______(5):
    print(i)
```

32. Get a slice from index 1 to 3 (not including 3):

```
letters = ["a", "b", "c", "d", "e"]
subset = letters[_______:_______]
```

33. Convert user input to an integer:

```
age = _______(input("Enter age: "))
```

34. Remove the last item from a list:

```
items = [1, 2, 3, 4]
items._______()
```

35. Check if a value is in a list:

```
numbers = [5, 10, 15]
if 10 _______ numbers:
    print("Found")
```

Part 3: Fix the Errors

36. Fix this code by crossing out the error and writing the correction:

```
user_age = "30"
next_year = user_age + 1
print(next_year)
```

37. Fix this code by crossing out the error and writing the correction:

```
name = "Alice
print(name)
```

38. Fix this code by crossing out the error and writing the correction:

```
student name = "Bob"
print(student name)
```

39. Fix this code by crossing out the error and writing the correction:

```
if score >= 60
    print("Pass")
```

40. Fix this code by crossing out the error and writing the correction:

```
numbers = [1, 2, 3]
print(numbers[3])
```

41. Fix this code by crossing out the error and writing the correction:

```
for i in range(3)
print(i)
```

42. Fix this code by crossing out the error and writing the correction:

```
fruits = ["apple", "banana"]
fruits.append "cherry"
```

43. Fix this code by crossing out the error and writing the correction:

```
if age >= 18:
print("Adult")
```

44. Fix this code so it prints the sorted list:

```
numbers = [15, 5, 10]
sorted_numbers = numbers.sort()
print(sorted_numbers)
```

45. Fix this code by crossing out the error and writing the correction:

```
word = "hello"
print(word.Upper())
```

Part 4: Code Reading & Output Prediction

46. What is the final value of count?

```
count = 0
for i in range(4):
    count = count + 2
print(count)
```

47. What will this print?

```
numbers = [10, 20, 30]
numbers.append(40)
print(len(numbers))
```

48. What is the output?

```
x = 15
if x > 10 and x < 20:
    print("Yes")
else:
    print("No")
```

49. What will this print?

```
text = "Python"
print(text[0] + text[-1])
```

50. What is the final value of total?

```
numbers = [5, 10, 15, 20]
total = 0
for num in numbers:
    total = total + num
print(total)
```